

28/03/19 -

Theorem - let $X \sim P_\theta$, $\theta \in \Theta$, let $W(x)$ be an unbiased estimator of $g(\theta)$ satisfying the condition required in CRLB. Then,
 $\forall \theta$ $\text{Var}_\theta(W(x))$ is equal to the CRLB if and only if $\exists$ a function $a(\theta)$ —

$$\text{s.t. } \forall \theta : \forall x : a(\theta) (W(x) - g(\theta)) = \frac{d}{d\theta} \log P_\theta(x)$$

Proof: In the derivation of CRLB check when Cauchy Schwartz become an equality —

$$(2) \text{Cov}(A, B)^2 \leq \text{Var}(A) \cdot \text{Var}(B)$$

$$\left(\sum_i a_i b_i \right)^2 \leq \sum_i a_i^2 \cdot \sum_i b_i^2$$

} vector form

equality is achieved if and only if vector a, b are scalar multiple of each other (i.e. in same direction)

$$(1) E[P_\theta]^2 \leq E P^2 \cdot E \theta^2$$

(1) equality is achieved if and only
if $P = \alpha Q$ for some $\alpha \in \mathbb{R}$

(62)

(ii) $A - E[A] = \alpha (B - E[B])$ for some $\alpha \in \mathbb{R}$

now proof: $A \equiv w(x)$
 $B \equiv \frac{\partial}{\partial \theta} \log P_\theta(x)$, under P_θ

therefore,

$$\alpha [w(x) - E_\theta[w(x)]] = \left(\frac{\partial}{\partial \theta} \log P_\theta(x) - E_\theta \left(\frac{\partial}{\partial \theta} \log P_\theta(x) \right) \right)$$

$$\alpha [w(x) - g(\theta)] = \frac{\partial}{\partial \theta} \log P_\theta(x)$$

hence,

$$\left\{ \begin{array}{l} \forall \theta \\ \forall x, \end{array} \alpha(\theta) [w(x) - g(\theta)] = \frac{\partial}{\partial \theta} \log P_\theta(x) \right\}$$

eg Unbiased estimation of σ^2 (without using / knowing μ).

$$\theta = \sigma^2 \quad x \equiv (x_1, \dots, x_n)$$

$$\frac{\partial}{\partial \theta} \log P_\theta(x) = \frac{\partial}{\partial \theta} \sum_{i=1}^n \log P_\theta(x_i)$$

$$= \frac{\partial}{\partial \theta} \sum_{i=1}^n \log \frac{1}{\sqrt{2\pi\theta}} e^{-(x_i - \mu)^2 / 2\theta}$$

$$= \sum_{i=1}^n \frac{\partial}{\partial \theta} \left\{ -\frac{1}{2} \log \theta - \frac{(x_i - \mu)^2}{2\theta} \right\}$$

$$= \sum_{i=1}^n \left\{ -\frac{1}{2\theta} + \frac{(x_i - \mu)^2}{2\theta^2} \right\}$$

how,

$$\sum_{i=1}^n \left\{ -\frac{1}{2\theta} + \frac{(x_i - \mu)^2}{2\theta^2} \right\} \stackrel{?}{=} a(\theta) \left(\underbrace{w(x_1, \dots, x_n)}_{\substack{\text{Cannot} \\ \text{dependent on } \mu, \theta}} - \theta \right) \quad (x_1, \dots, x_n)$$

$$-\frac{n}{2\theta} + \frac{1}{2\theta^2} \sum_{i=1}^n (x_i^2 - 2\mu x_i + \mu^2)$$

Cannot
dependent on μ, θ

Can depend on
 μ —
but can't depend
on $(x_1, \dots, x_n)$

It follows that this identity in
 $x_1, \dots, x_n, \theta, \mu$, cannot be satisfied,
unless $w(x_1, \dots, x_n)$ depend on μ .

In fact, this identity holds with

$$w(x_1, \dots, x_n) = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

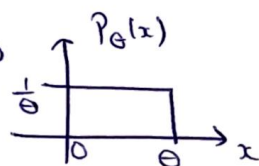
ie estimator
depends
on μ .

— unbiased
estimator for σ^2

→ Hence, the CRLB for unbiased estimation of (μ, σ^2)
cannot be achieved.

Remark: There are estimation problem where CRLB techn
conditions are not satisfied, but where the
RHS of the CRLB can be beaten with
natural estimators —

e.g. $(x_1, \dots, x_n) \stackrel{iid}{\sim} \text{Unit}(0, \theta)$, $\theta > 0$



Since we can't interchange $\div$

(G3)

$$\frac{d}{d\theta} \int_0^\theta T(x) P_\theta(x) dx$$

$\hat{\theta} = \max(X_1, \dots, X_n)$ not unbiased estimator of θ .

$\hat{\theta} = \frac{(n+1)}{n} \max(X_1, \dots, X_n)$ ~~is~~ unbiased estimator of θ

$w(x) =$

and gets lesser variance than RHS of CRLB.

Sufficient Statistics :-

estimation problem $x \sim P_\theta$, θ -unknown

$w(\cdot)$ designed to estimate / approximate θ .

∅ A sufficient statistics is a function $T(x)$ that "preserves" all essential information about θ .

∅ A sufficient statistics for θ is a function $T(\cdot)$ st.

$\forall \theta$: The prob. distribution of x given $T(x)$ does not depend on θ i.e

$$\left. \begin{array}{l} P_{x \sim P_\theta} [X=x | T(x)=t] \text{ must not} \\ \text{depend on } \theta \end{array} \right| \begin{array}{l} \forall x \\ \forall \theta \\ \forall t \end{array}$$

Note:

$$\boxed{\theta \rightarrow x \rightarrow T(x)}$$

so, $(\theta, x, T(x))$ form a Markov chain i.e

$$P[T(x)|x, \theta] = P[T(x)|x]$$

$T(x)$ being a suff. statshcs for $\theta \Leftrightarrow$

$(\theta, T(x), x)$ is a Markov chain

$$\text{i.e. } P(x|\theta, T(x)) = P(x)T(x)$$

example 1) $X \sim P_\theta$

$T(x) = x$ is a (trivial) sufficient statistics.

$$\begin{aligned} \forall \theta \\ \forall x : P_\theta [X=x|T(x)=t] \\ \forall t \end{aligned} = P_\theta [X=x|X=t] = \begin{cases} 1 & x=t \\ 0 & x \neq t \end{cases}$$

does not depend on θ

2) sufficient statshcs for the Bernoulli distribution

$X = (X_1, \dots, X_n) \stackrel{\text{i.i.d.}}{\sim} \text{Ber}(\theta)$, $0 \leq \theta \leq 1$

$$T(x) = \sum_{i=1}^n X_i$$

$$\begin{aligned} \forall \theta \\ \forall x : P_\theta [X=x|T(x)=t] \\ \forall t \end{aligned}$$

$$= P[(X_1, \dots, X_n) = (x_1, \dots, x_n) | \sum_i x_i = t]$$

$$\swarrow \text{if } t \neq \sum_i x_i$$

$$= 0$$

$$= P_{\theta} \left[X_1 = x_1, X_2 = x_2, \dots, X_n = x_n, \sum_i X_i = \sum_i x_i \right] \quad (64)$$

$$P_{\theta} \left[\sum_i X_i = \sum_i x_i \right]$$

$$= \frac{\theta^t (1-\theta)^{n-t}}{\binom{n}{t} \theta^t (1-\theta)^{n-t}}$$

$$= \underline{\underline{\text{does not upon } \theta}}$$

example: Sufficient statistics for Normal density —
 $X \equiv (X_1, \dots, X_n) \sim N(\mu, \sigma^2)$
 $\theta \equiv \text{unknown}$

$T(X) \equiv (X_1, \dots, X_n)$ — not sufficient statistics
 (loses μ)

$$\rightarrow \pi(x) \equiv \left(\frac{1}{n} \sum_{i=1}^n X_i, \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{n-1} \right)$$

$$\equiv \left(\sum_{i=1}^n X_i, \sum_{i=1}^n (X_i - \bar{X})^2 \right)$$

is a sufficient statistics for (μ, σ^2) .